LITERATURE REVIEW

Social media is nowadays one of the main news sources for millions of people around the globe due to their low cost, easy access, and rapid dissemination. This however comes at the cost of dubious trustworthiness and significant risk of exposure to 'fake news', intentionally written to mislead the readers. Automatically detecting fake news poses challenges that defy existing content-based analysis approaches. One of the main reasons is that often the interpretation of the news requires the knowledge of political or social context or 'common sense', which current natural language processing algorithms are still missing. Recent studies have empirically shown that fake and real news spread differently on social media, forming propagation patterns that could be harnessed for the automatic fake news detection. Propagation-based approaches have multiple advantages compared to their content-based counterparts, among which is language independence and better resilience to adversarial attacks. In this paper, we show a novel automatic fake news detection model based on geometric deep learning. The underlying core algorithms are a generalization of classical convectional neural networks to graphs, allowing the fusion of heterogeneous data such as content, user profile and activity, social graph, and news propagation. Our model was trained and tested on news stories, verified by professional fact-checking organizations, that were spread on Twitter. Our experiments indicate that social network structure and propagation are important features allowing highly accurate (92.7% ROC AUC) fake news detection. Second, we observe that fake news can be reliably detected at an early stage, after just a few hours of propagation. Third, we test the aging of our model on training and testing data separated in time. Our results point to the promise of propagation-based approaches for fake news detection as an alternative or complementary strategy to content-based approaches.

Paper Title	Objective	Methods Used	Tools Used	Advantages	Limitations
Fake News Detection on Social Media: A Data Mining Perspective	To study techniques for detecting fake news on social media	Data mining, feature extraction, classification algorithms	ML models, social media datasets	Comprehensive overview of fake news detection methods	Does not focus on real-time implementation
BERT: Pre-training of Deep Bidirectional Transformers	To improve NLP tasks using deep learning models	Transformer-based deep learning, contextual embeddings	BERT, NLP frameworks	High accuracy in text understanding and classification	Requires high computational resources
Fake News Detection on Twitter	To detect fake news specifically on Twitter platform	NLP techniques, machine learning classification	Twitter datasets, ML tools	Focused on real-world social media data	Performance depends on dataset quality
FakeNewsTracker: Collection, Detection and Visualization	To collect, detect, and visualize fake news data	Data collection, visualization, classification methods	Web tools, ML models	Provides end-to-end system approach	Complex system design